Upcoming + Reminders

Find all links on the **course website:** cogs9.github.io/fa26

📚 **Project:** *Checkpoint 1* DUE: today)

All due at 11:59pm

📚 **Read:** *Reading(s) 4, Data Visualization*

- Reading quiz 4 (DUE: Wednesday Week 6)

Guest Lecture — Week 8

- Participation is worth 1 point
  - Fill out a Google Form of your takeaways
  - Interactive Q&A!
- **Microsoft AI** generates tools like Github Copilot
- Jolene also worked (with me) at **Reddit**

[recorded] Week 8

AI and Industry Data Science

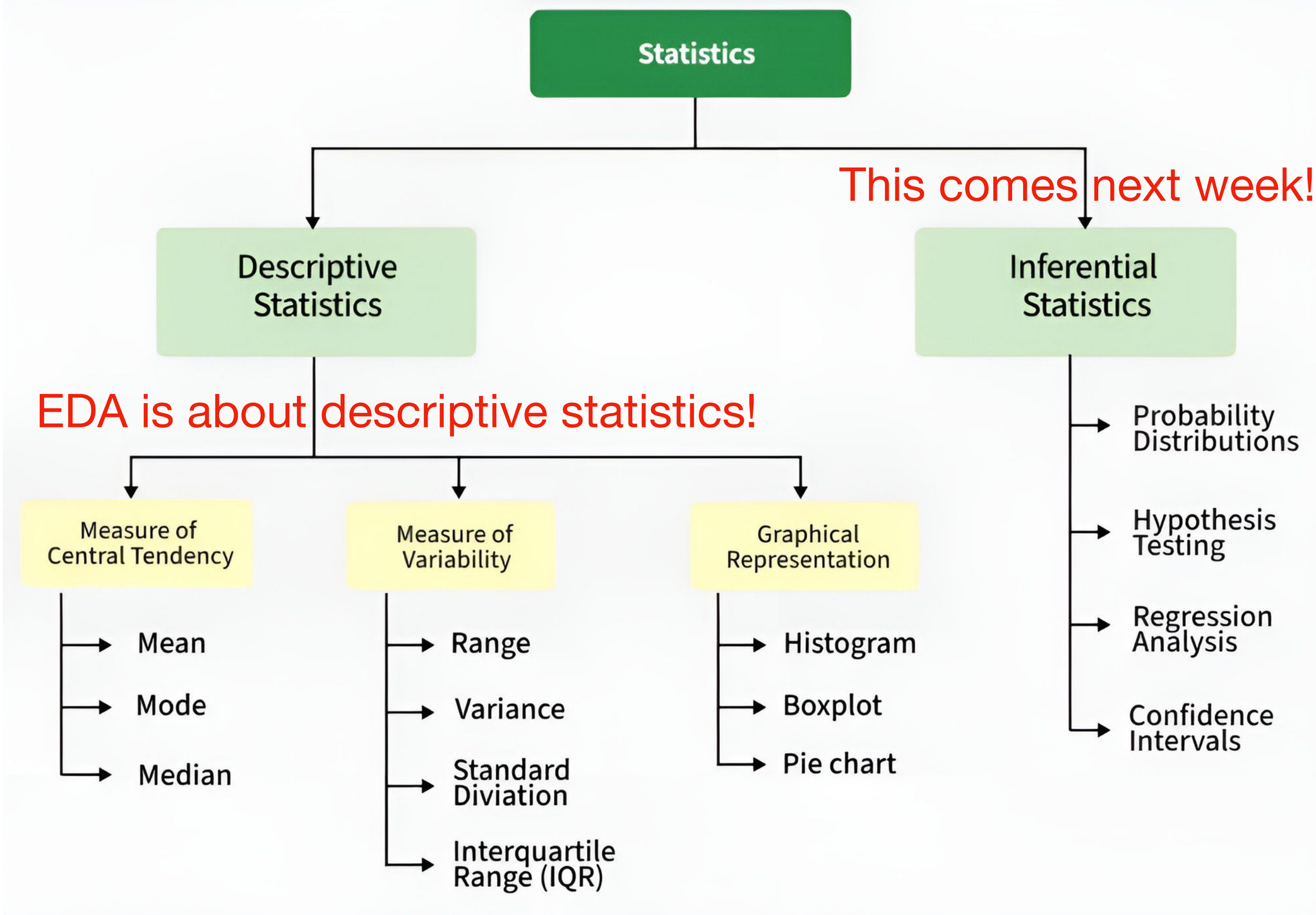


A. Jolene Mork

cogs 9 | Week 5

EDA & Statistical Testing

Brendan Tomoschuk


Statistics
This comes next week!
Descriptive Statistics
Inferential Statistics
EDA is about descriptive statistics!
Measure of Central Tendency
Measure of Variability
Graphical Representation
Mean
Mode
Median
Range
Variance
Standard Diviation
Interquartile Range (IQR)
Histogram
Boxplot
Pie chart
Probability Distributions
Hypothesis Testing
Regression Analysis
Confidence Intervals

Why do we do EDA?

Main purpose: To understand your dataset before further analysis and modeling!

- **Step 1:** Understand the data
- **Step 2:** Univariate EDA (1 variable)
- **Step 3:** Bivariate EDA (2 variables)
- **Step 4:** Multi-variate EDA (2+ variables)

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course code:
WSTAPX

Why is it important to do univariate analysis? *Select all that apply.*

(**A**) It proves that one variable causes changes in another variable

(**B**) It helps detect outliers or weird values

(**C**) It reveals a variable's distribution/spread

(**D**) It helps determine which models and statistical tests are appropriate

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WSTAPX

Why is it important to do univariate analysis? *Select all that apply.*

(A) It proves that one variable causes changes in another variable

(B) It helps detect outliers or weird values

(C) It reveals a variable's distribution/spread

(D) It helps determine which models and statistical tests are appropriate

Step 3: Bivariate EDA

Main goal: Understanding relationships or comparisons between two variables

- Quantitative-quantitative relationships
- Categorical-quantitative relationships
- Categorical–categorical relationships

Why is this important?:

- Detecting associations or trends between variables
  - Influences analysis or modeling methods: Models may depend on linear relationships between variables, or one variable might influence the relationship between two other variables (interaction effect)
- Comparing groups: *Is Group A different than Group B in some way?*
- Helps generate hypotheses: *“Group A differs from Group B” or “X predicts Y.”*

Step 3: Bivariate EDA

Quantitative-quantitative relationship

Key visual: scatterplots!

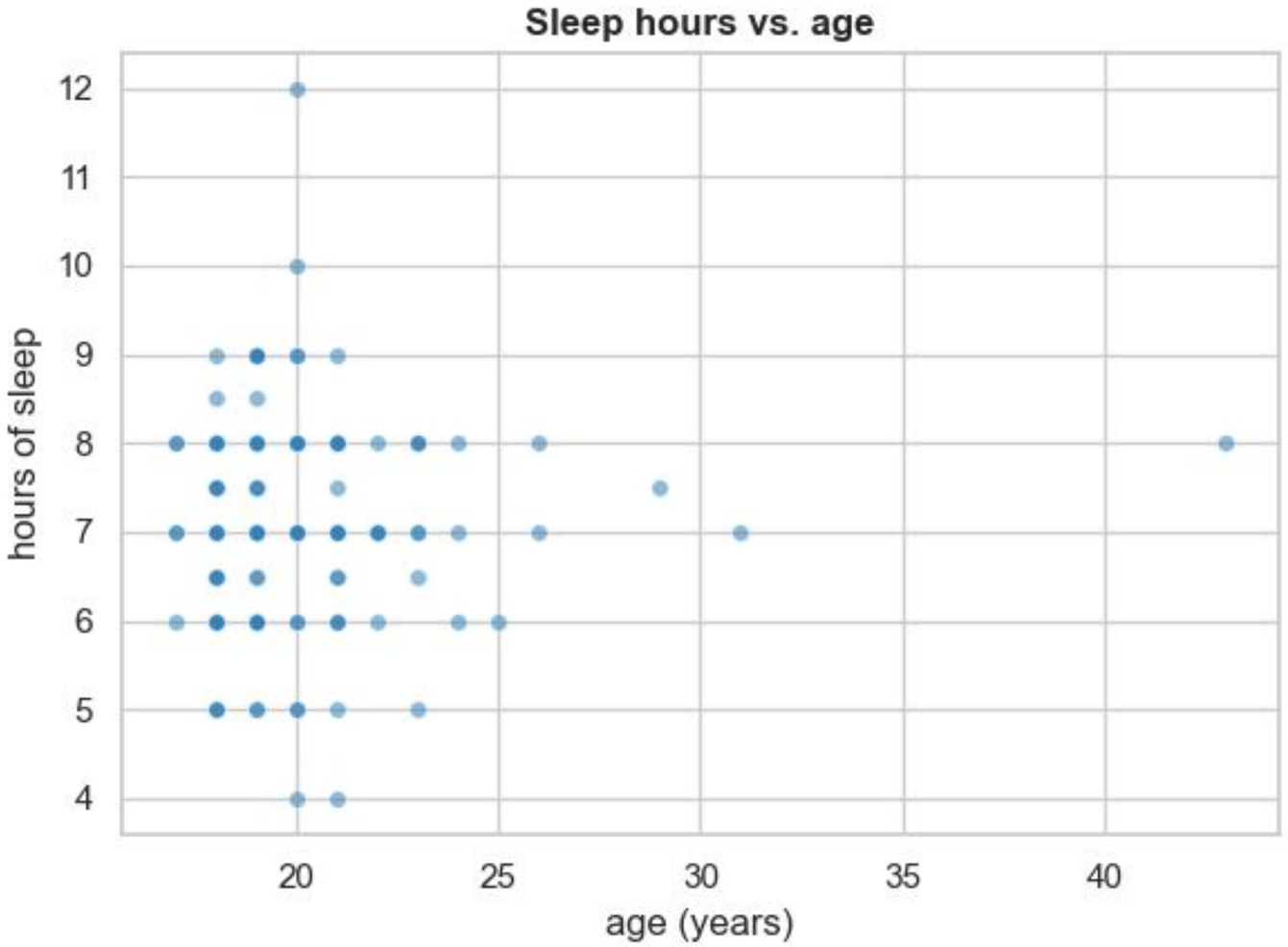
- Detecting associations or trends between quantitative variables

Questions to ask:

Is there a relationship between sleep and age?

Is is linear or non-linear?

Is it positive or negative? Or neither?



Step 3: Bivariate EDA

Quantitative-quantitative relationships

Key visual: scatterplots!

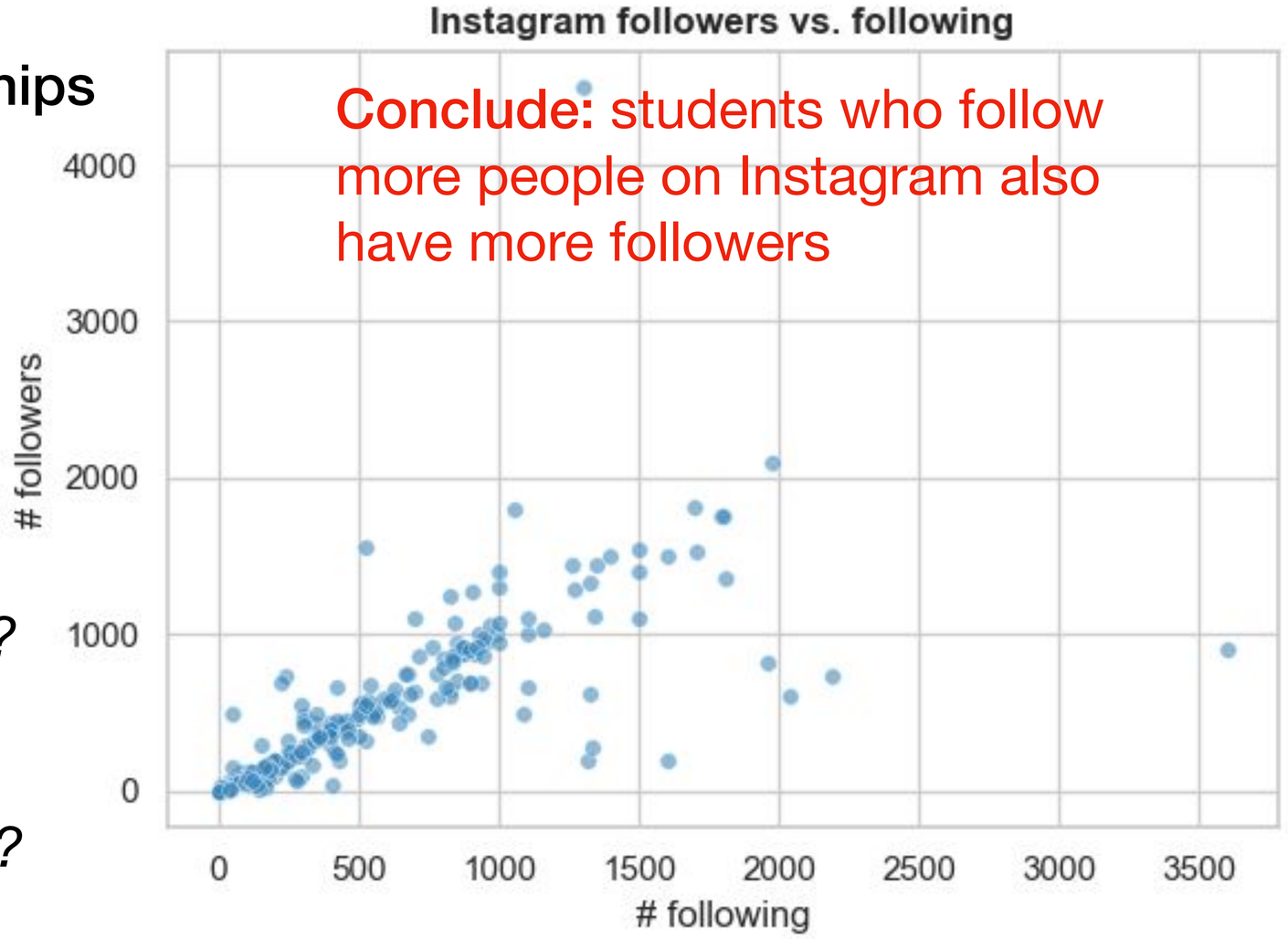
- Detecting associations or trends between quantitative variables

Questions to ask:

Is there a relationship between followers and how many you follow?

Is is linear or non-linear?

Is it positive or negative? Or neither?



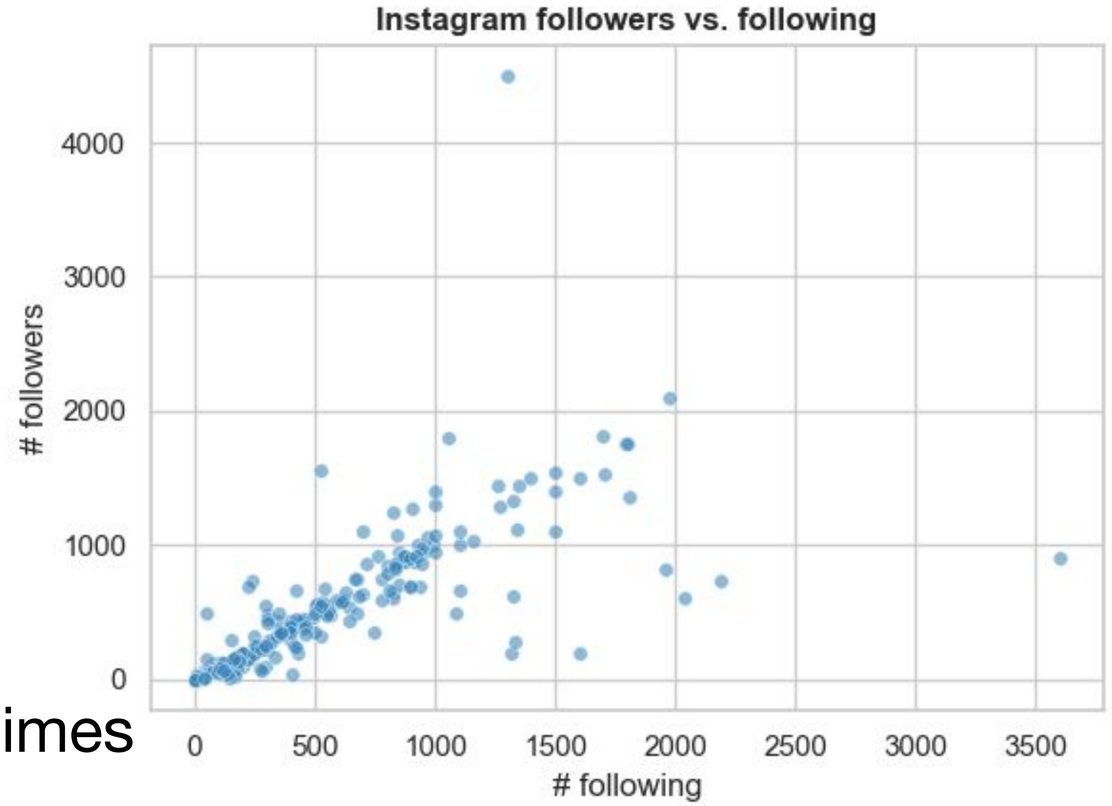
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course code:
WSTAPX

Does this plot show that following more people on Instagram causes you to have more Instagram followers?



(A) Yes

(B) No

(C) Sometimes

(D) I wish

(E) I'm not sure / I'm confused

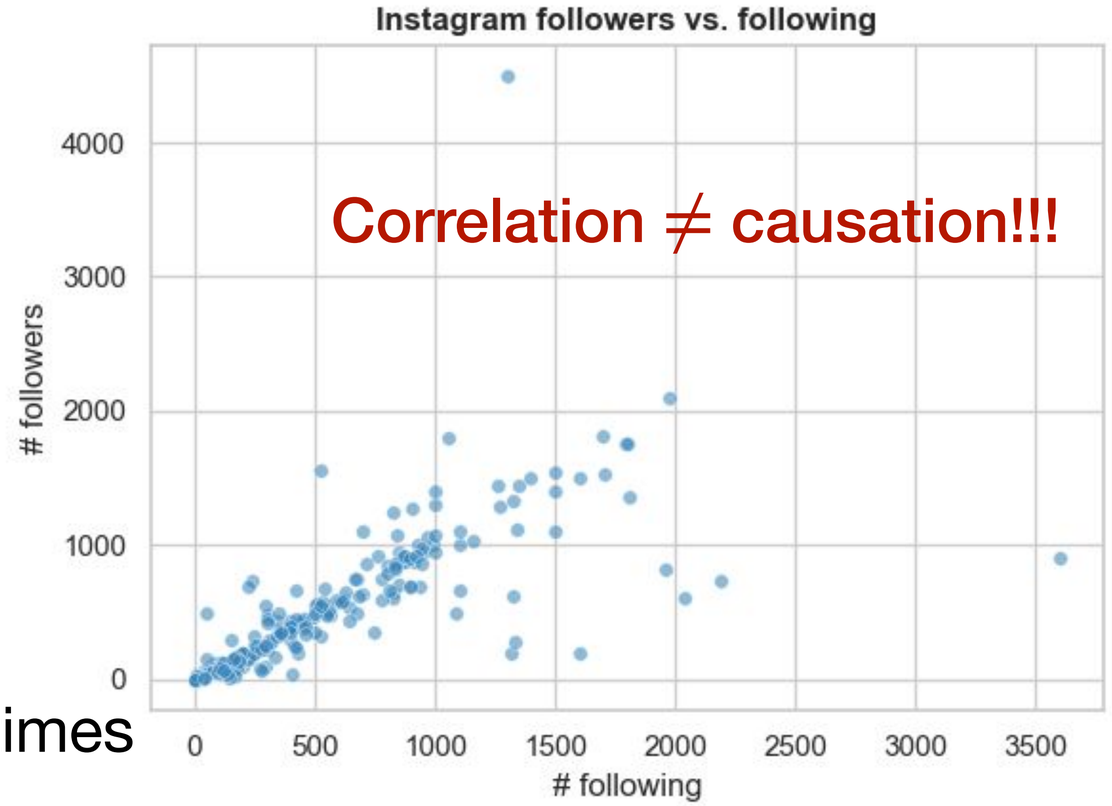
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course code:
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Step 3: Bivariate EDA

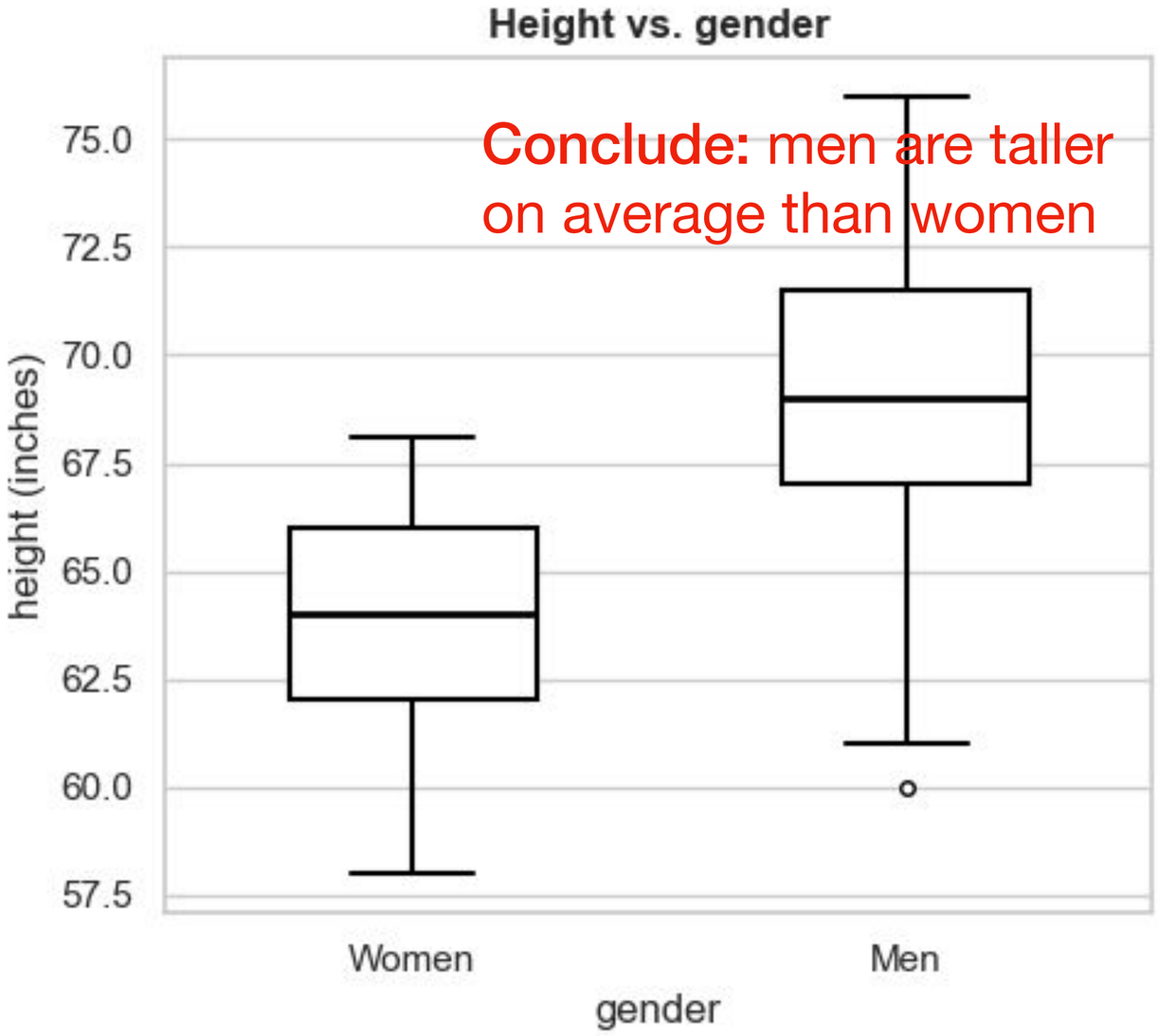
Categorical-quantitative relationships

Key visual: side-by-side box plots

- Comparing a quantitative variable across categories

EDA question:

Is student height related to self-reported gender identity?



Step 3: Bivariate EDA

Categorical-quantitative relationships

Key visual: side-by-side box plots

- Comparing a quantitative variable across categories

EDA question:

Is student age related to class standing (year)?

Age vs. Year



Step 3: Bivariate EDA

Categorical-categorical relationships

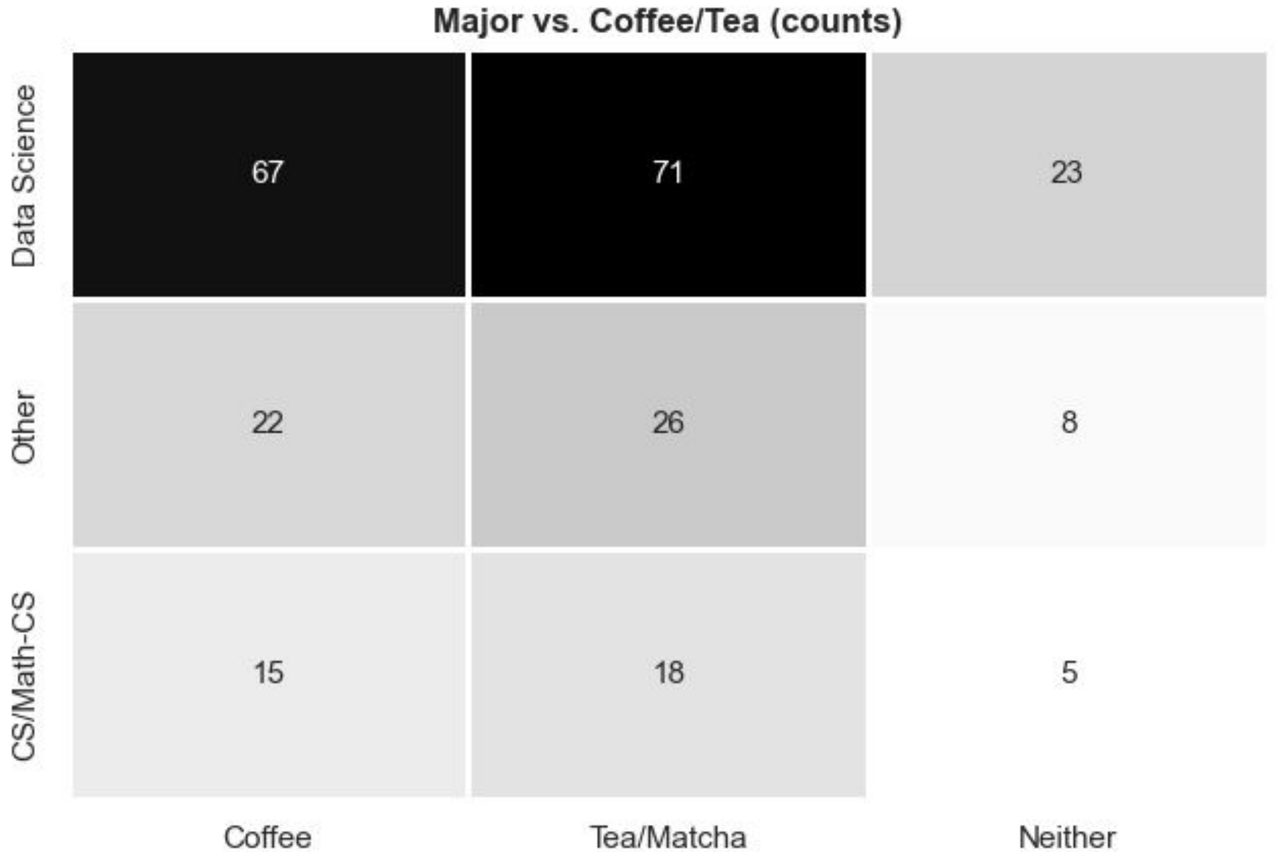
Key visual: contingency tables
(+ heat map)

- Detecting associations or trends between categorical variables

EDA question:

Is student major related to whether they prefer coffee or tea/matcha? (Or neither?)

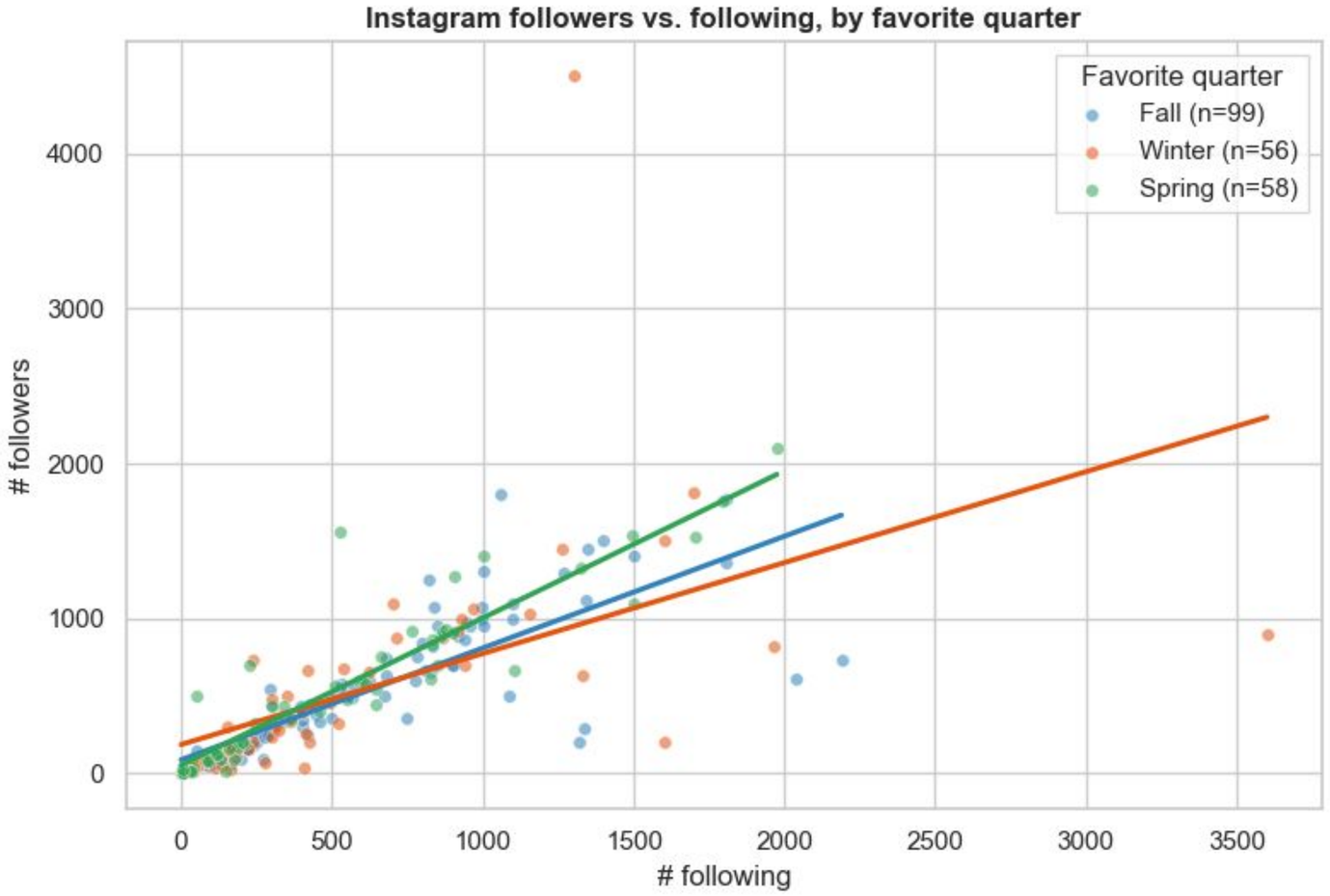
Conclude: All majors like tea/matcha more than coffee (to different, small degrees)



Step 4: Multivariate EDA

Exploring relationships between 2+ variables

- Exploring how the association of two quantitative variables varies across categories
- **Example:** *Is the relationship between followers and following dependent on a student's reported favorite quarter?*



Conclude: If you love spring quarter, you have a stronger relationship between followers and following?

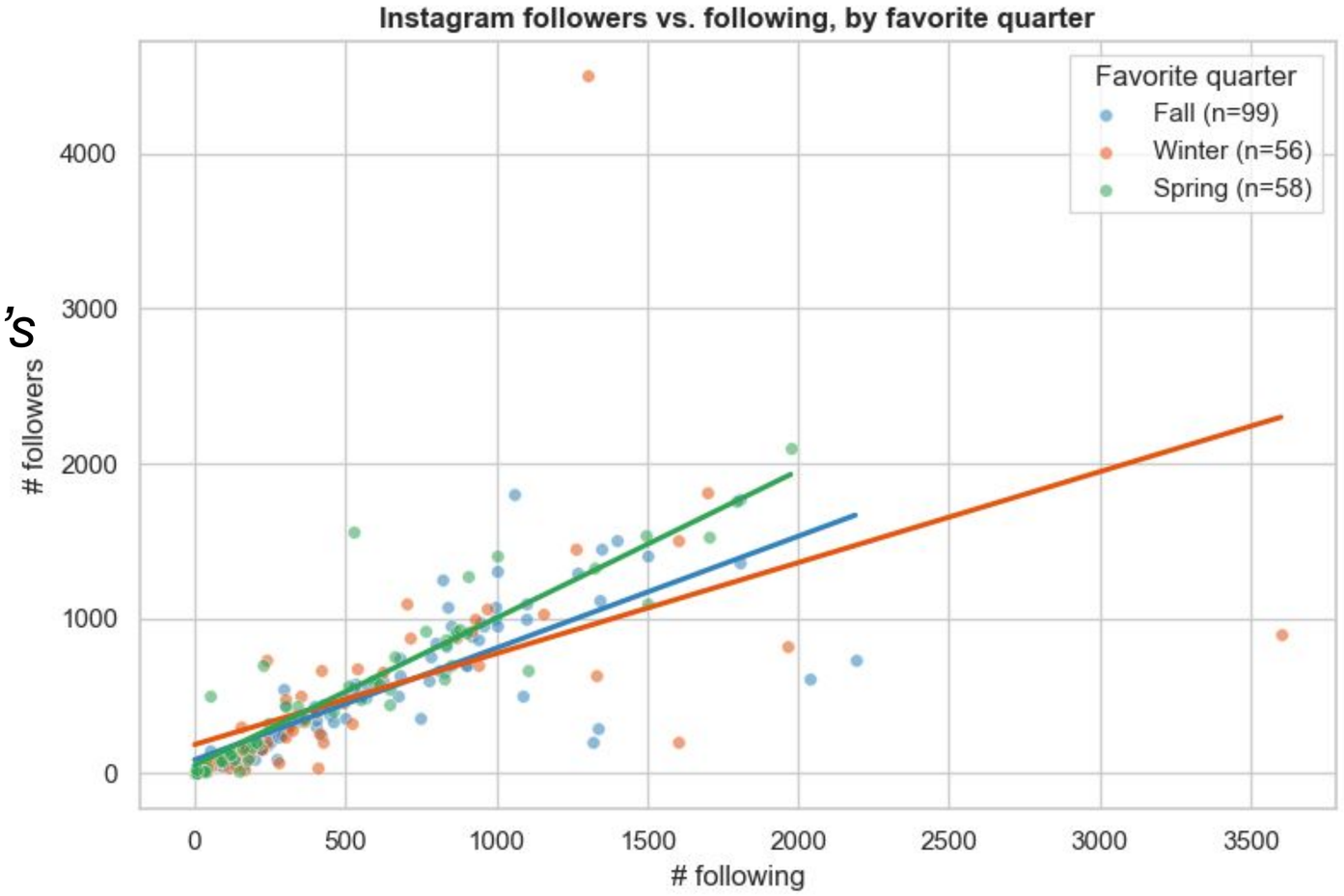
Step 4: Multivariate EDA

Exploring relationships between 2+ variables

- *Is the relationship between followers and following dependent on a student's reported favorite quarter?*

What are the issues here?

- No *a priori* reason for relationship
- Slightly imbalanced classes?
- Outliers?



Discussion (~5 mins)

Groups of 2-3:

Data science question: *Which variables are most predictive of whether a student will attend lecture or not in Week 5 of COGS 9, Spring 2026?*

1. Describe the EDA plot(s) you would make to explore the data with respect to this question (name the variables + specify plot type)
    - Univariate EDA
    - Bivariate EDA
2. What would you learn from this plot?

Discussion (~5 mins)

Data science question: *Which variables are most predictive of whether a student will attend lecture or not in Week 5 of COGS 9, Spring 2026?*

Qualitative

- First name
- Last name
- Year
- Preferred pronouns (optional)
- What is your major?
- Do you plan to attend lectures?
- What's your natural hair color?
- How do you describe your race and/or ethnicity? (Select all that apply)
- Where were you born?
- What's your favorite quarter?
- Do you prefer T/Th or M/W/F classes?

Quantitative

- Coffee or tea/matcha?
- Cats or dogs?
- If money wasn't an object, what would you want to be when you grow up?
- What is your height in inches?
- How old are you?
- How many siblings do you have?
- How many hours of sleep do you get a night on average?
- What is your spotify listening age?
- How many instagram followers do you have?
- How many people do you follow on instagram?

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course code:

WSTAPX

Which of the following visualizations is best for comparing the distribution of a quantitative variable across categories?

(A) Scatterplot

(B) Side-by-side box plots

(C) Two histograms

(D) Side-by-side bar plots

(E) Both B and D

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Which of the following visualizations is best for comparing the distribution of a quantitative variable across categories?

(A) Scatterplot

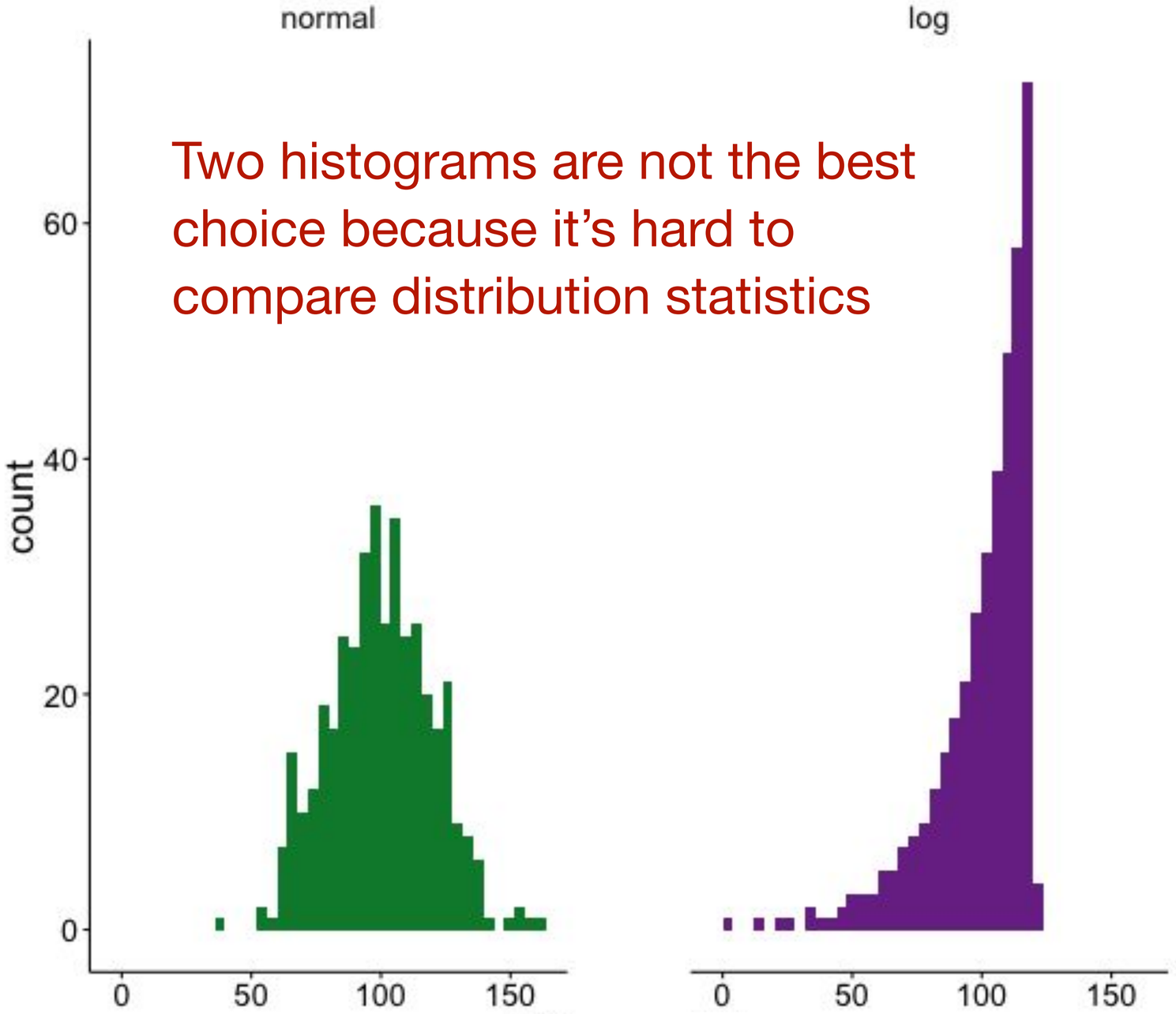
(B) Side-by-side box plots → best for comparing the **distribution** of a quantitative variable across categories

(C) Two histograms

(D) Side-by-side bar plots → best for comparing **group means** across categories

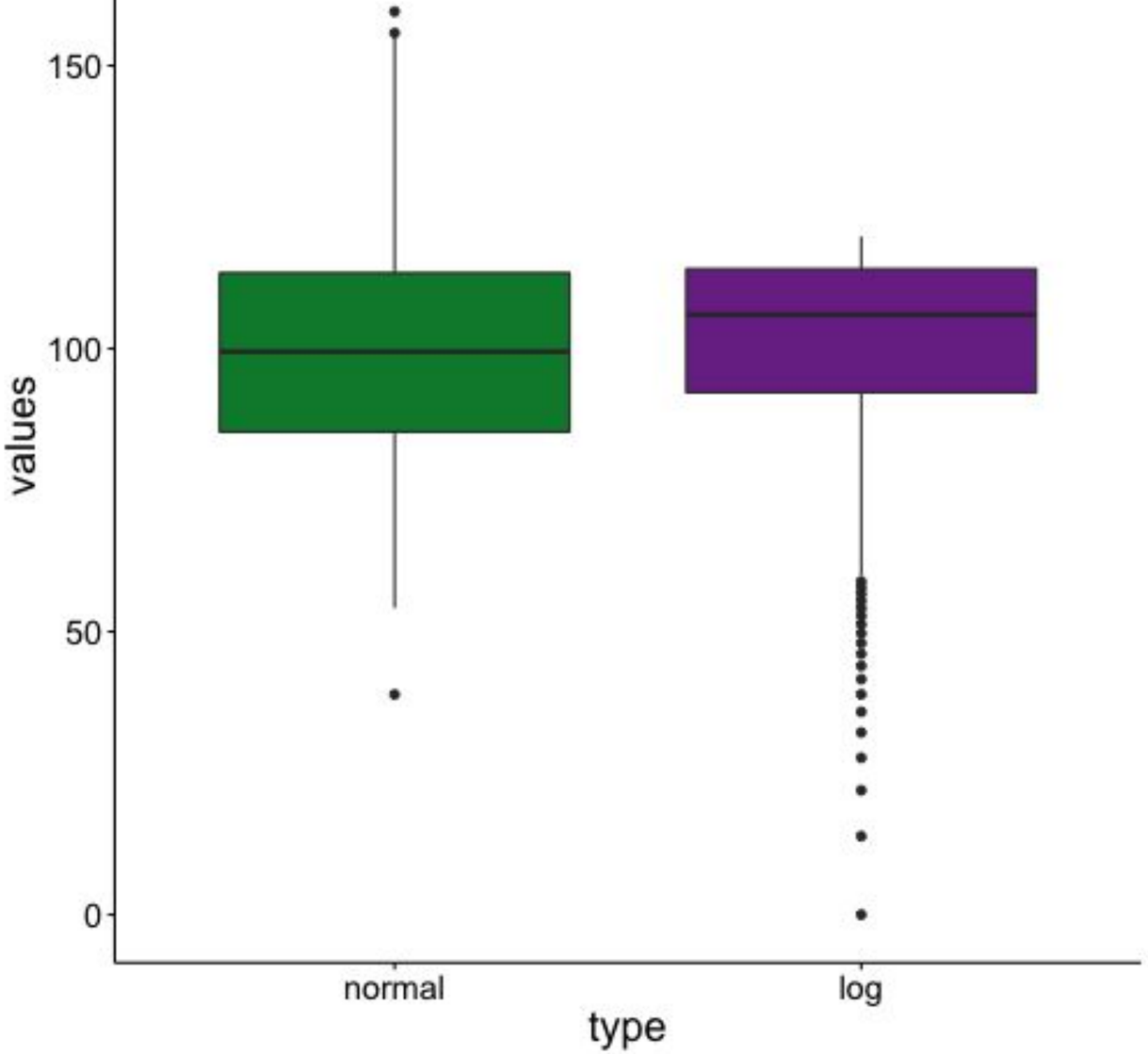
(E) Both B and D

Imagine two distributions with different skew:

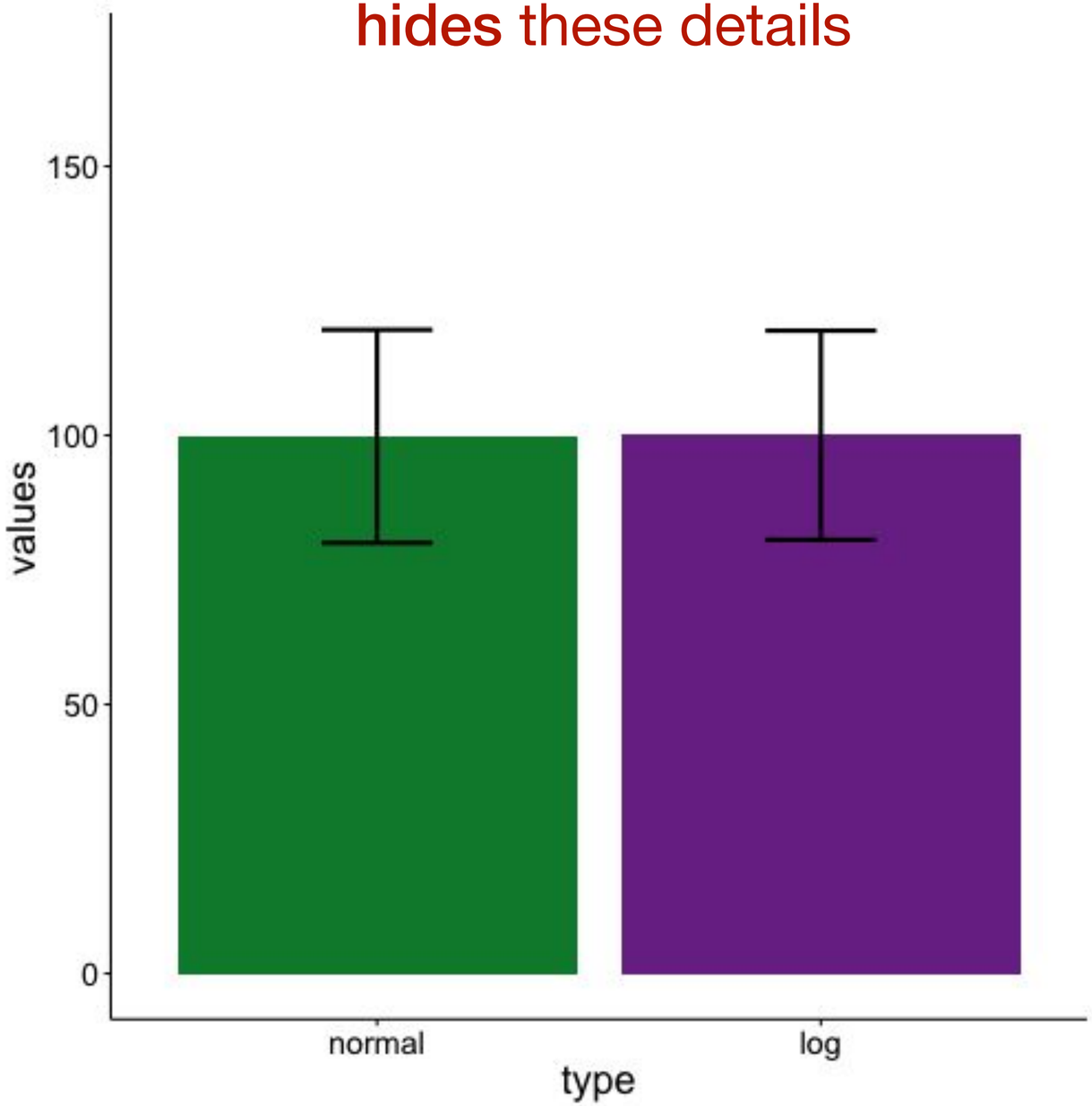


Two histograms are not the best choice because it's hard to compare distribution statistics

Box plots can show the **data distribution** (spread, skew, outliers) within each group



A bar plot of means **hides** these details



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course code:
WSTAPX

What kind of bivariate relationship does a contingency table show?

(A) Quantitative-quantitative relationship

(B) Categorical-quantitative relationship

(C) Categorical-categorical relationship

(D) Both A and C

(E) I'm not sure / I'm confused

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course code:

WSTAPX

What kind of bivariate relationship does a contingency table show?

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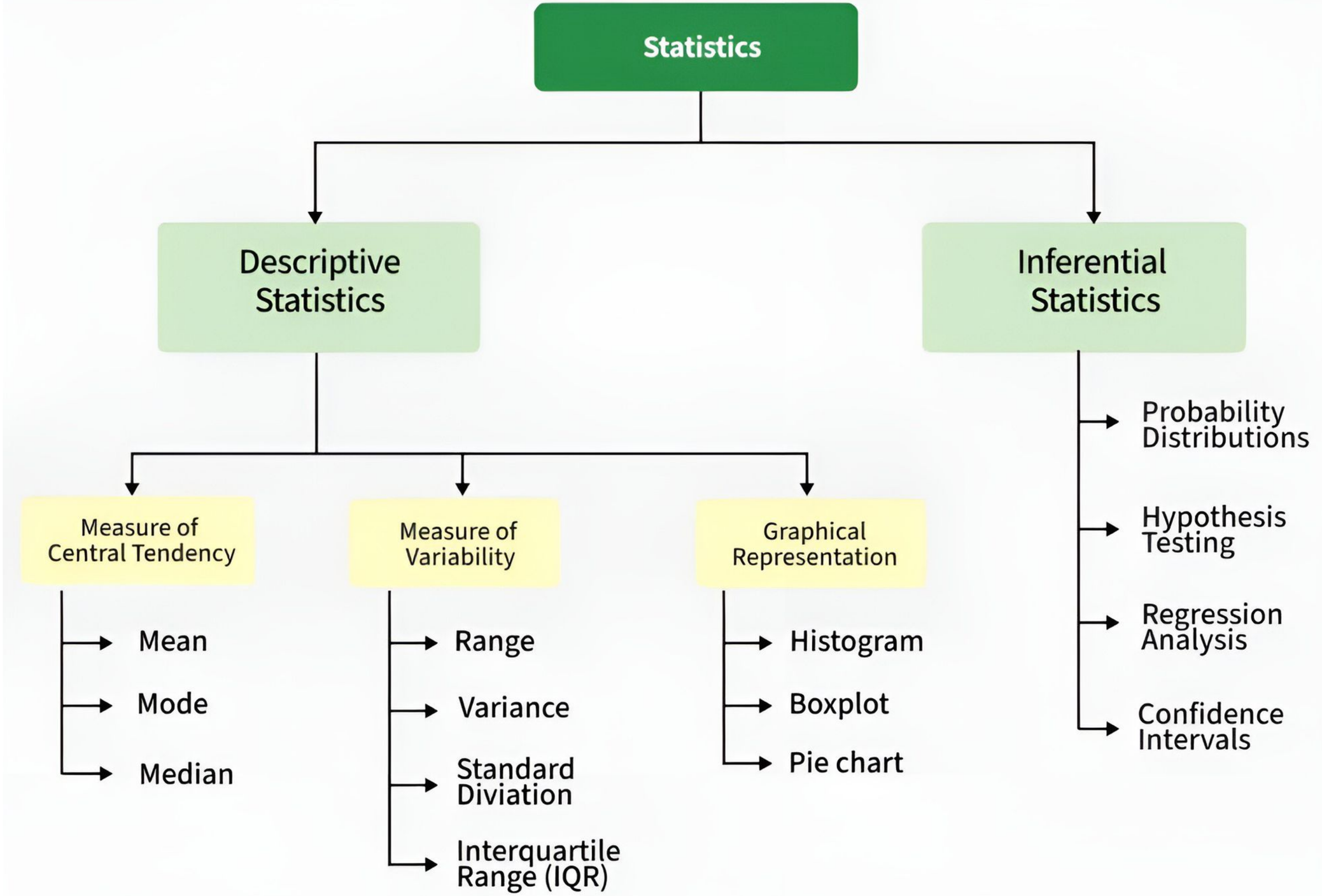
(D) Both A and C

(E) I'm not sure / I'm confused

Statistical Inference!

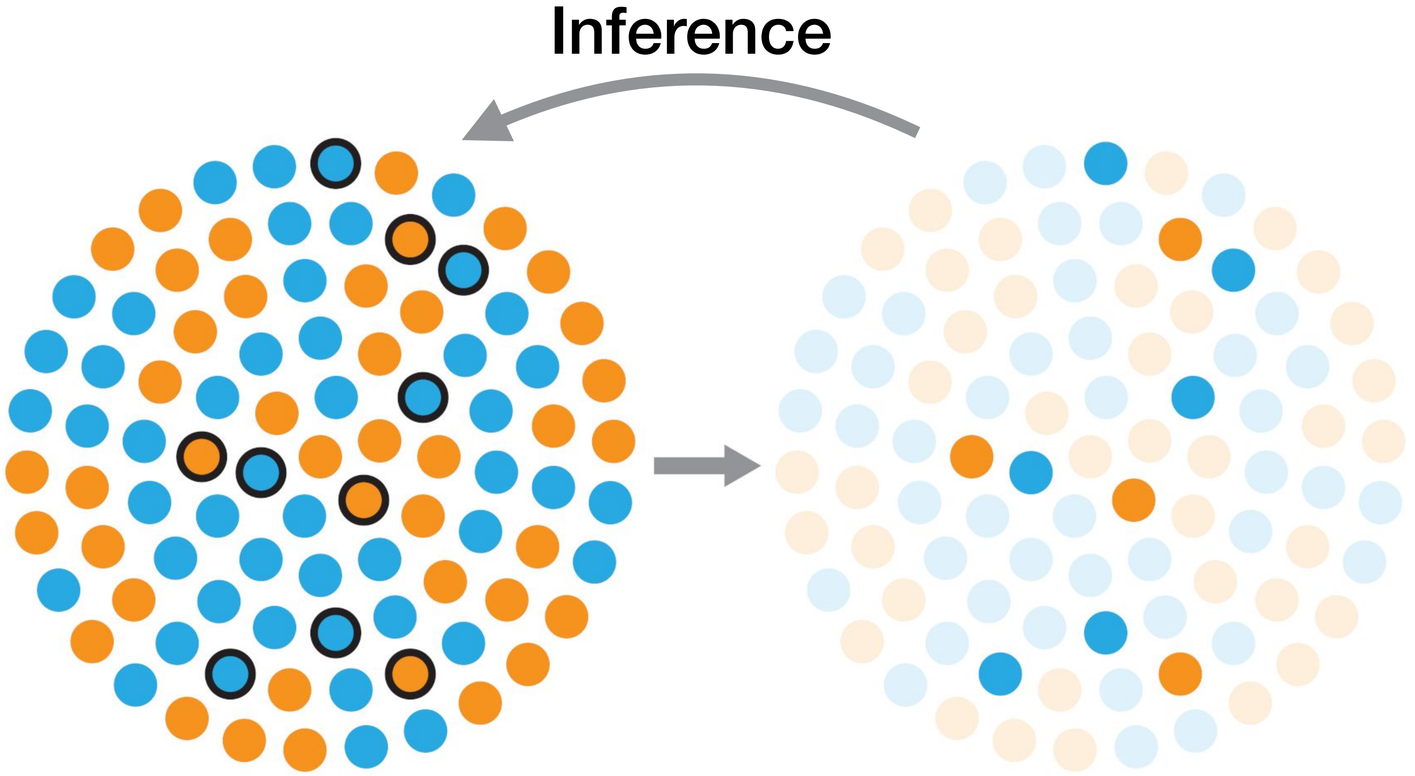
Learning Goals

1. Explain the goal of statistical inference and define statistical inference terms: *population, sample, parameter, statistic, and sampling distribution*
2. Quantify uncertainty in parameter estimates using standard error and confidence intervals
3. Formulate a testable hypothesis and choose an appropriate statistical test
4. Interpret the results of statistical tests and corresponding p-values


Statistics
Descriptive Statistics
Inferential Statistics
Measure of Central Tendency
Measure of Variability
Graphical Representation
Mean
Mode
Median
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Variance
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Interquartile Range (IQR)
Histogram
Boxplot
Pie chart
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Confidence Intervals

Statistical Inference

- Statistical inference is the process of drawing conclusions about a population from a sample of data



Population

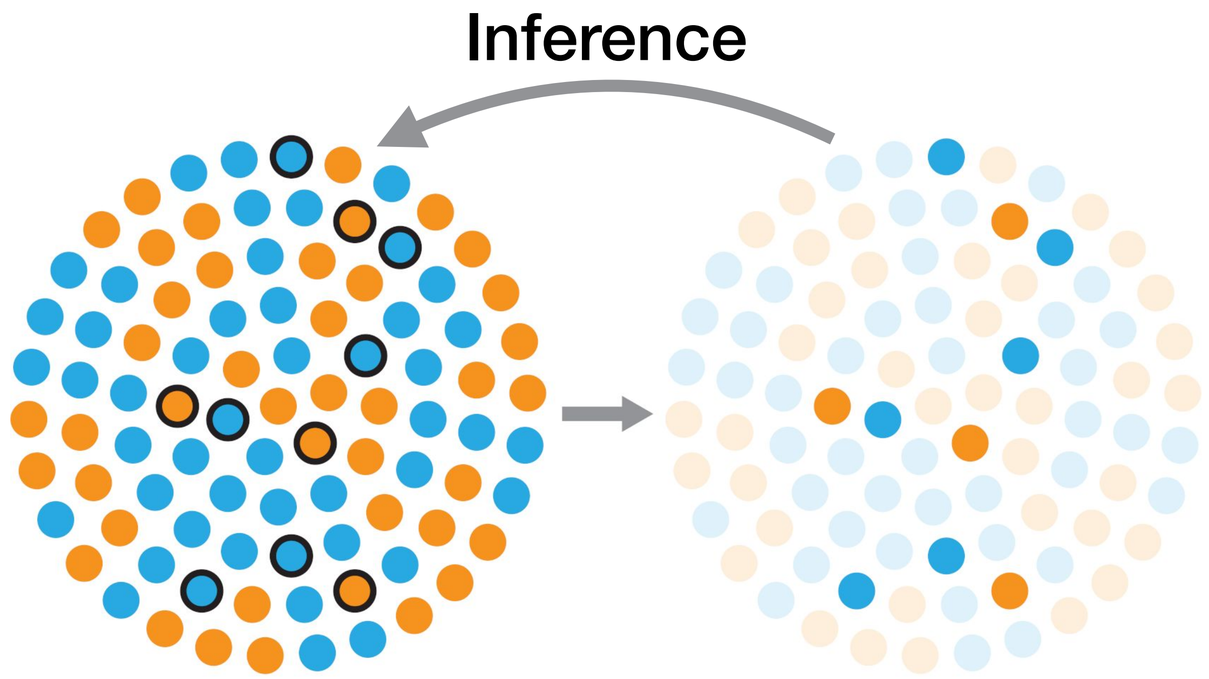
All students in COGS 9

Sample

A subset of students in COGS 9

Statistical Inference

- Statistical inference is the process of drawing conclusions about a **population** from a **sample** of data
- Statistical inference uses *sample statistics* to estimate *population parameters*



Population

A parameter is a characteristic of the population

(e.g., the *population* mean, std, variance)

Sample

A statistic is a characteristic of the sample

(e.g., the *sample* mean, std, variance)

Types of statistical inference

- **Univariate inference:** estimate a single population parameter (e.g., μ, σ) about a single variable
- **Bivariate inference:** draw conclusions about the relationship between two variables (comparison or association)
  - Quantitative ↔ quantitative: test association in values
  - Categorical ↔ quantitative: difference in means across groups
  - Categorical ↔ categorical: test association or difference in proportions
- **Multivariate inference:** drawing conclusions about relationships between variables while accounting for other variables; detect confounding and interaction effects *(study this in a later course)*

Taking a sample of the COGS 9 dataset

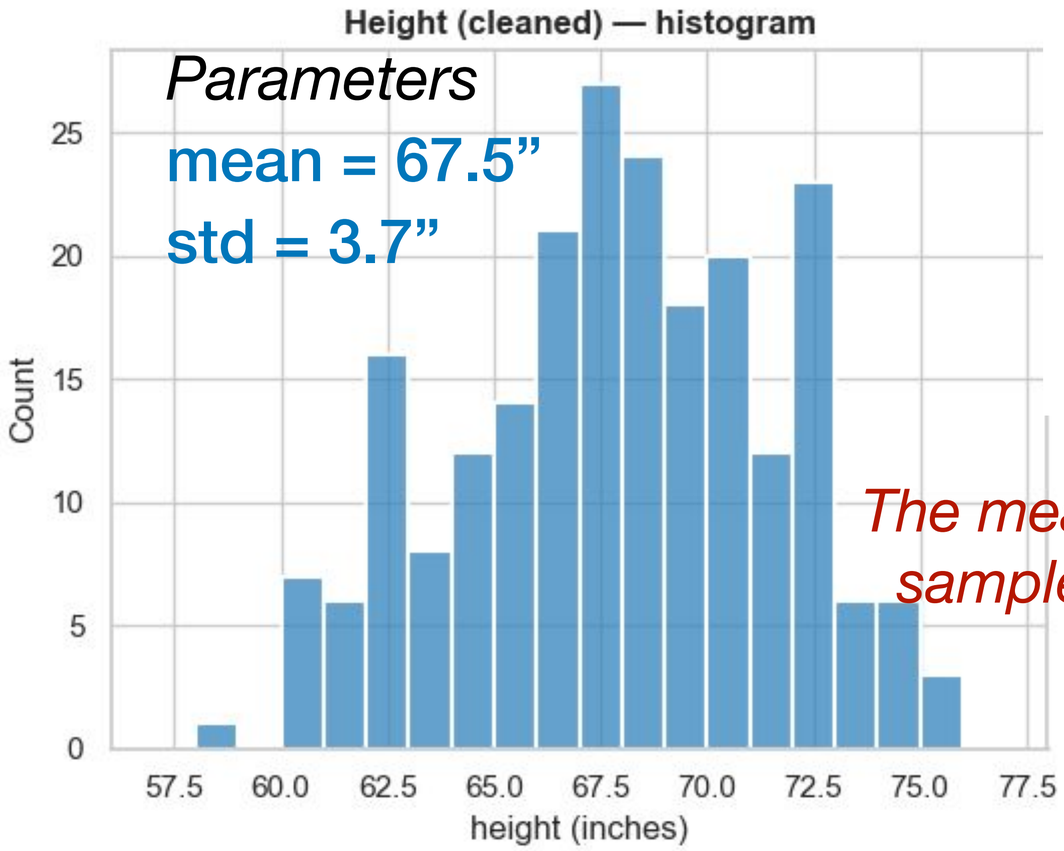
Year	Preferred pronouns (optional)	What is your major?	Do you plan to attend lectures?	What is your height in inches?	How old are you?	What's your natural hair color?	How do you describe your race and/or ethnicity? (Select all that apply)	Where were you born?	How many siblings do you have?	How many hours of sleep do you get a night on average?	What is your spotify listening age?	How many instagram followers do you have?	How many people do you follow on instagram?	What's your favorite quarter?	Do you prefer...	Coffee or tea/matcha?	Cats or dogs?	If money wasn't an object, what would you want to be when you grow up?
Third-year	he/him	Data science	Yes	69.0	22.0	Black	Asian	North or Central America	1.0	10.0	24.0	485.0	740.0	Fall	Tues/Thurs classes	Neither	Dogs	Someone who can help people
Second-year	he/him	Data science	Sometimes	65.0	19.0	Black	Asian, Indian	Asia	0.0	6.0	74.0	898.0	1107.0	Spring	Tues/Thurs classes	Coffee	Dogs	A professional billiards player
Second-year	she/her	Other	No	62.0	19.0	Black	Asian	North or Central America	2.0	7.0	NaN	568.0	327.0	Spring	Tues/Thurs classes	Neither	Dogs	An entrupreneur
First-year	he/him	Other	Yes	74.0	19.0	Black	Asian	Asia	1.0	9.0	0.0	76.0	105.0	Fall	Tues/Thurs classes	Tea/Matcha	Cats	capybara
First-year	she/her	Cognitive science	Yes	NaN	18.0	Brown	Asian	Asia	1.0	5.0	NaN	69.0	72.0	Winter	Tues/Thurs classes	Coffee	Cats	A marine biologist
First-year	he/him	Other	Yes	6.0	18.0	Brown	White	North or Central America	0.0	8.0	50.0	160.0	300.0	Fall	Mon/Wed/Fri classes	Neither	Cats	Writer
Second-year	she/her	Cognitive science	Yes	67.0	20.0	Brown	White	North or Central America	1.0	7.0	65.0	378.0	353.0	Spring	Mon/Wed/Fri classes	Coffee	Dogs	Music teacher
First-year	he/him	Data science	Yes	65.0	18.0	Black	Black or African American	North or Central America	1.0	7.0	62.0	115.0	170.0	Fall	Mon/Wed/Fri classes	Coffee	Cats	NaN

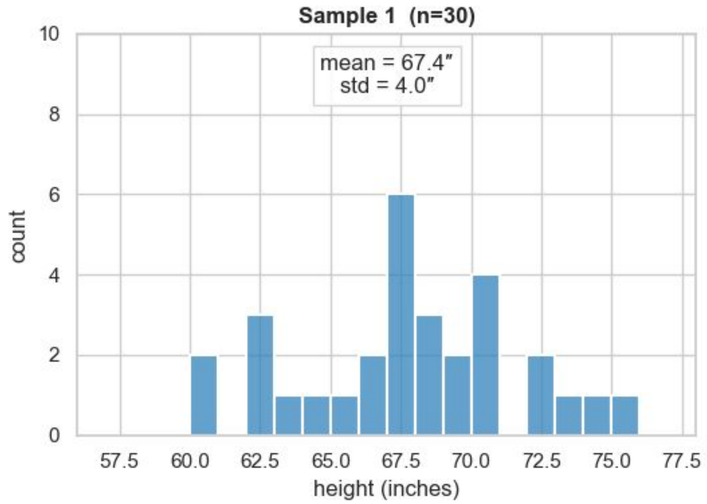
Univariate statistical inference

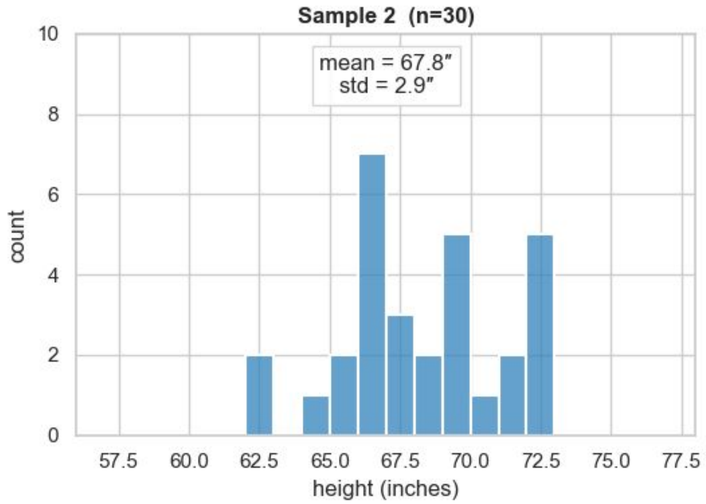
- **Univariate inference:** estimate a single population parameter (e.g., μ mean, σ SD, median, p proportion) about a single variable
  - **Inference tools:** standard error, confidence intervals

Example: Sample of COGS 9 heights

Population: n = 224

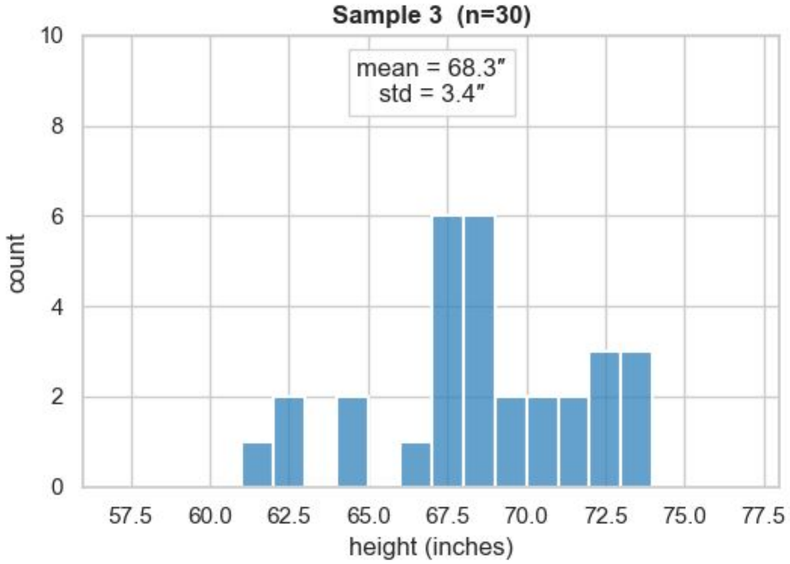






The means vary from sample to sample!

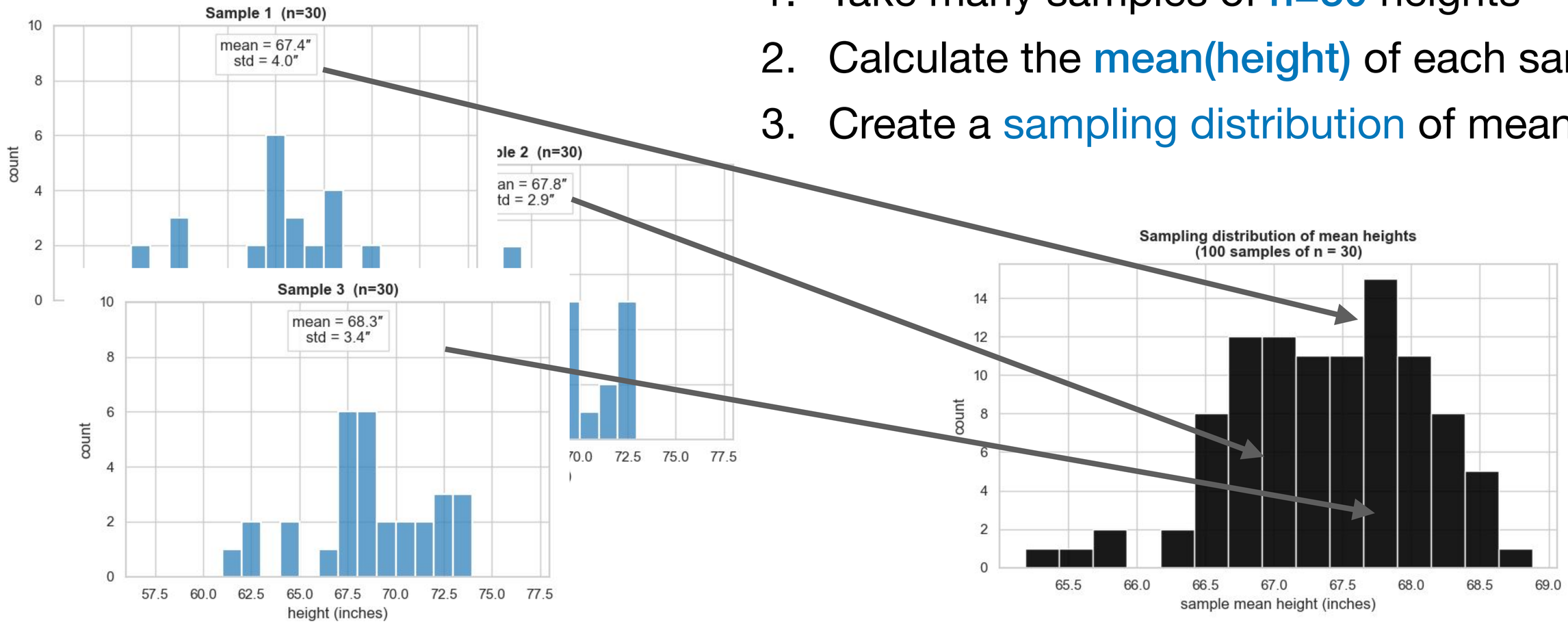
Samples:
n = 30



Example: Sample of COGS 9 heights

The **sampling distribution** is the distribution of a statistic over many samples:

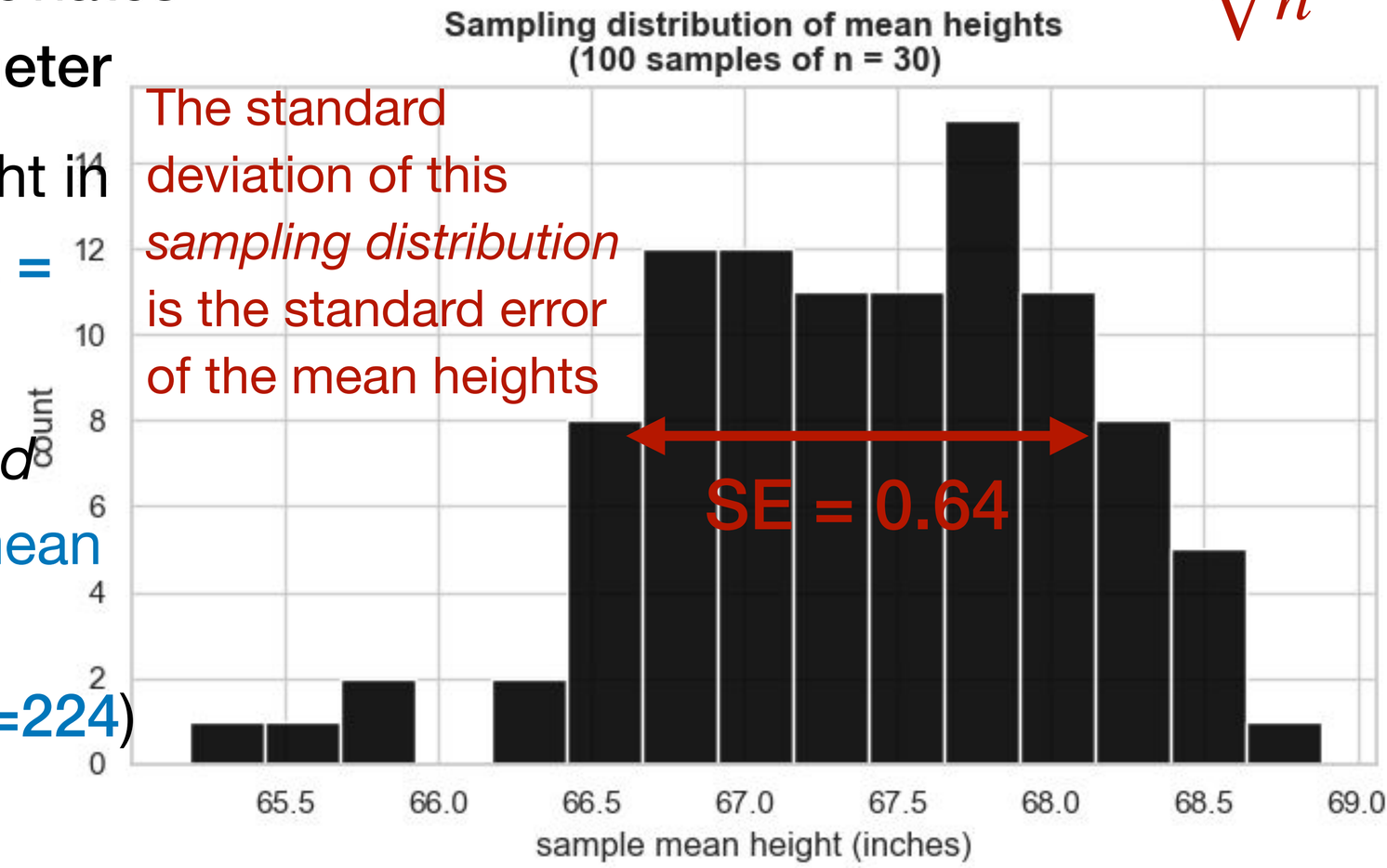
1. Take many samples of **n=30** heights
2. Calculate the **mean(height)** of each sample
3. Create a **sampling distribution** of mean heights



Quantifying uncertainty of a sample statistic

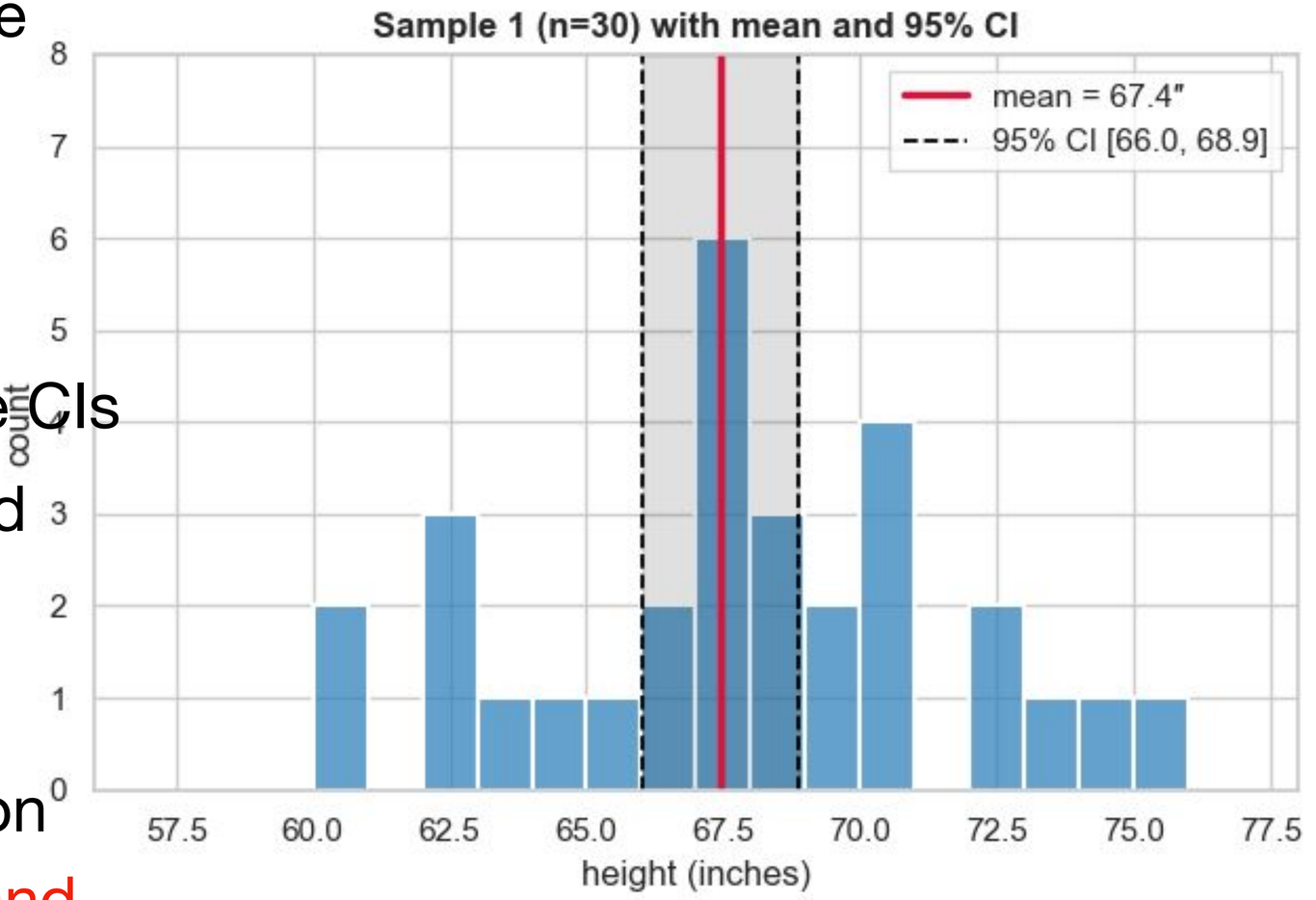
- The **standard error (SE)** quantifies how much a sample **statistic** deviates from the true population **parameter**
- ***Report:*** The sample mean height in COGS 9 is **66.3 inches** with **SE = 0.52 inches**
- This means that *across repeated random samples,* the sample mean height (**n=30**) differs from the true population mean height (**n=224**) by about **0.64 inches**

$$SE = \frac{s}{\sqrt{n}}$$



Quantifying uncertainty of a sample statistic

- A **confidence interval (CI)** uses the standard error to give a **range of plausible values** for the true parameter
- Across many samples, 95% of the CIs $[\text{mean} \pm 1.96 \times \text{SE(mean)}]$ would capture the true *population* mean height
- ***Report:*** We estimate the population mean height to be between 66.0 and 68.9 inches (95% CI)



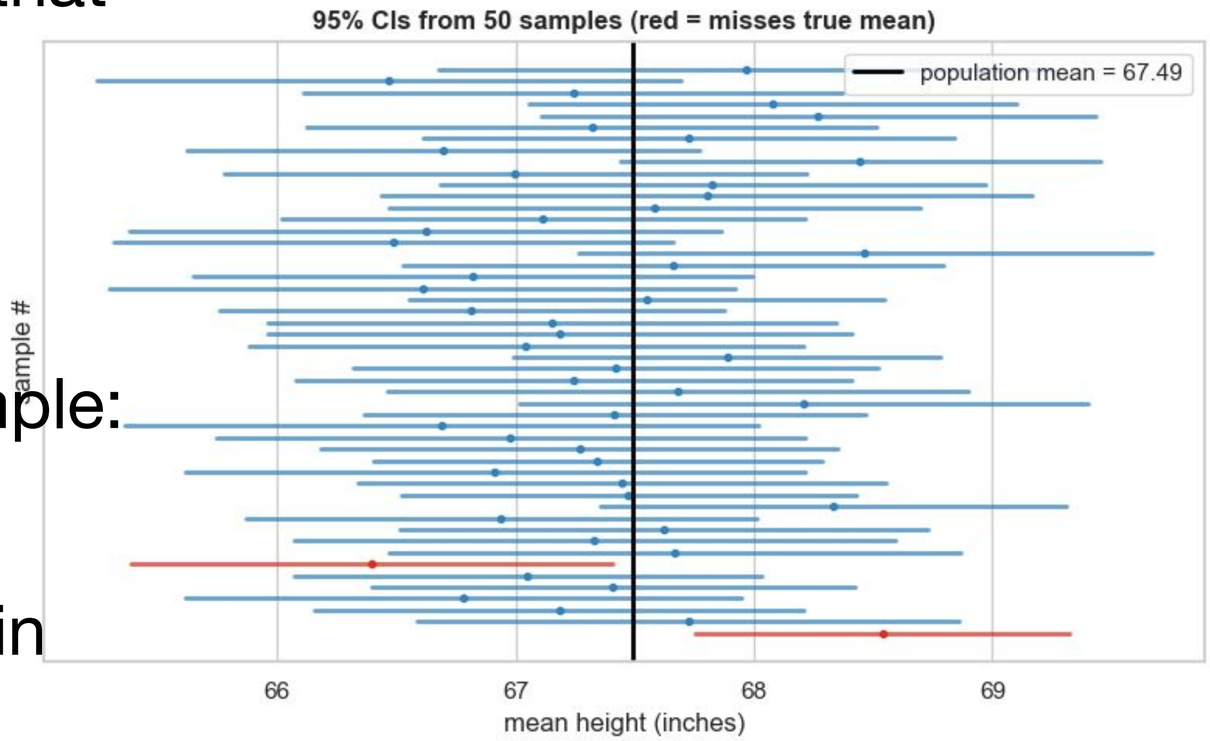
Quantifying uncertainty of a sample statistic

A 95% confidence interval (CI) means that if you…

1. Take **100 samples** from the population
2. Calculate a 95% CI for each sample:

   $[\text{mean} \pm 1.96 \times \text{SE(mean)}]$

- 95 out of those 100 CIs would contain the true population mean height
- 5 out of those 100 CIs would NOT contain the true population mean height



heights in COGS 9

Unlike this COGS 9 dataset, we DON'T always have access to the entire population!



Population

You have only a *fraction* of that data to answer your question



Sample

8 billion people in the world
342 million people in the USA
3 billion Instagram users

Statistical inference is important for knowing how we can *generalize* the conclusions from our sample to the population